

# Sam Stryker

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## SUMMARY

Incoming University of Chicago student with advanced preparation in physics, mathematics, and computer science, pursuing original research in quantum error mitigation in collaboration with a research scientist at IBM T.J. Watson Research Center. Recognized with a 3rd place finish in Computer Science at Regeneron WESEF research competition and both regional and national academic awards. Experience includes hardware design and detector construction, alongside independent projects in RF systems and software. In parallel, service as a certified volunteer firefighter and EMT has strengthened teamwork, resilience, and leadership under pressure. Studying physics and molecular engineering with a focus on quantum engineering, with the goal of advancing quantum technology to the point of possible widespread adoption.

## EDUCATION

### The University of Chicago, Chicago, IL

Class of 2030 (matriculating September 2026)

#### Information and Coursework:

- B.S. in Physics with Molecular Engineering double major (Quantum Engineering Track)
- Pathways in Molecular Engineering credit-bearing summer class - Grade: 100%

### Sungkyunkwan University, Seoul, South Korea

June 2026 - July 2026

#### Information and Coursework:

- Took Statistics in Python and Korean during the International Summer Semester
- Selected to give semester graduation speech

### Horace Greeley High School, Chappaqua, NY

September 2022 - June 2026

#### Relevant Coursework: GPA: 4.0 / 4.0

- AP Physics C (Calculus-based Mechanics and E&M), AP Chemistry, AP Calculus BC
- Advanced Topics in College Mathematics (Calculus III + Linear Algebra)
- Advanced Topics in College Computer Science (Advanced Java data structures and algorithms)

#### Awards:

- Summa Cum Laude; NYS Attorney General's Triple "C" Award; AP Scholar with Distinction x2; National Top School Recognition Award

## RESEARCH EXPERIENCE

### CERN Beamline for Schools Project Lead

2026

- Developed an experimental proposal to investigate radiation-induced changes in the quantum properties of nitrogen-vacancy centers in CVD diamond using CERN's T9 beamline.
- Organized team, found faculty sponsor, led research question development, and wrote the final research proposal.
- Project endorsed by The Meriles Group at CCNY - resulted in offer of research position within the group for Summer 2027.

### Independent Researcher (affiliated with IBM T.J. Watson Research Center)

2023 - 2025

- Phase I: Bayesian ZNE with Calibration-Informed Noise Models (2023-2024)
  - Developed novel Bayesian-enhanced Zero-Noise Extrapolation (ZNE) methods for quantum error mitigation
  - Built custom calibration-based noise models for IBM quantum computers; achieved 87% error reduction
  - Drafting manuscript for submission to Quantum / IEEE Quantum
  - [github.com/samstryker/Bayesian-ZNE-PhaseI](https://github.com/samstryker/Bayesian-ZNE-PhaseI)

#### Awards:

- 3rd Place - Computer Science | Regeneron WESEF
  - Largest regional ISEF-affiliated high school research competition in the U.S.
- 1st Place - Computer Science | Somers Science Research Fair
  - Largest high school research proposal competition in the U.S.
- Phase II: Bayesian ZNE with Quantum Shadow Priors (2024 - 2025)
  - Designing hybrid Bayesian-ZNE framework with quantum shadows for scalable error mitigation in resource-constrained quantum systems
  - Benchmarking methodology on molecular simulation circuits (H2 and LiH VQE) running on IBM Quantum hardware

## WORK EXPERIENCE

### Firefighter - Chappaqua Volunteer Fire Department, Chappaqua, NY

January 2024 - Present

- Certifications: Firefighter I, CPR (Adult/Pediatric), Hazardous Materials First Responder, Exterior/Interior Firefighting Operations
- Respond to emergency fire and rescue calls requiring immediate decision-making under extreme pressure and life-threatening conditions
- Operate specialized equipment and coordinate team-based operations in critical emergency scenarios
- Demonstrate reliability, discipline, and leadership in high-stakes situations

### Emergency Medical Technician (EMT) - Chappaqua EMS + Pleasantville EMS

December 2025 - Present

- Respond to 911 calls as a New York State-certified EMT, completing agency field clearance
- Assist the crew chief with patient assessment, vital signs, basic life support, and patient transport
- Prepare and operate medical equipment during cardiac, trauma, and medical emergencies
- Coordinate with other EMTs, paramedics, firefighters, and other first-response personnel

### Mathematics Instructor - Mathnasium

December 2024 - January 2026

- Teaching K-12 students mathematics from elementary concepts through calculus

- Develop adaptive instructional methods for individual learning needs and cognitive differences
- Improved ability to communicate technical concepts to diverse audiences
- Entrusted with the supervision and care of children of all ages

## RELEVANT TECHNICAL PROJECTS

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### RF Security & Wireless Protocol Analysis

2023 - Present

- Designed and constructed rolljam device to analyze automotive rolling-code cryptography vulnerabilities and wireless security weaknesses
- Conducted comprehensive protocol analysis using software-defined radio (SDR) platforms and WiFi analysis tools
- Investigated security vulnerabilities in WiFi, NFC, radio frequency, and RFID communication protocols

### Scintillation-Based Cosmic-Ray Muon Detector

June, 2025

- Fabricated high-energy particle detection system using BC-408 scintillator and silicon photomultiplier readout
- Custom-designed and surface-mount soldered PCB for signal processing and peak detection

### Equation-Based Image Rendering Tool

May, 2024

- Built a program in Java that converts input images to graphable equation sets
- Utilized multiple data structures and algorithms

## OTHER HONORS, AWARDS, AND CERTIFICATIONS

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- NSLI-Y Scholarship Recipient (Korean) - U.S. Department of State
  - Scholarship to study in-demand languages necessary for national defense
  - 7-week scholarship to study Korean in South Korea - Most competitive region
- IBM Basics of Quantum Information Badge
- IBM Fundamentals of Quantum Algorithms Badge

## SKILLS

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Computational - Python (Qiskit, NumPy, SciPy, Pandas, Matplotlib), Java (data structures, algorithms, object-oriented programming), Linux/Bash (Ubuntu/Debian), Git/GitHub (version control), Fusion 360, AutoCAD  
Quantum Computing - IBM Qiskit, Variational quantum algorithms (VQE), Quantum error mitigation (ZNE, Bayesian methods)